

# PART IV

## USING OOP IN GAME DEVELOPMENT

In this part of the book, we'll build some sample games using `pygwidgets`. I'll also introduce the `pyghelpers` module, which includes a number of classes and functions that can be useful in building game programs.

Chapter 12 revisits the Higher or Lower game from Chapter 1. We'll build a version of the game with a graphical user interface, and I'll introduce `Deck` and `Card` classes that can be reused in any card game program.

Chapter 13 focuses on timers. We'll build a number of different timer classes that allow your program to keep running while concurrently checking for a certain time limit.

Chapter 14 discusses different animation classes you can use to show sequences of images. This will allow you to easily build more artistic games and programs.

Chapter 15 introduces an approach to building a program that can contain many scenes, like a start scene, a play scene, and a game over scene. I'll show a `SceneMgr` class that is designed to manage any number of programmer-built scenes, and we'll use it to build a Rock, Paper, Scissors game.

Chapter 16 demonstrates how to show and react to different types of dialog boxes. You'll then use everything you've learned to build a fully functioning animated game.

Chapter 17 introduces the concept of design patterns, using the model, view, controller pattern as an example. It then provides a short wrap-up for